

Model Comparison Report

Introduction

This report compares the performance of the deep learning models developed for multi-class skin disease classification. All models were trained using the same preprocessed dataset and evaluated using identical performance metrics to identify the best-performing model.

Evaluation Metrics

The models were evaluated using the following metrics:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC
- Loss

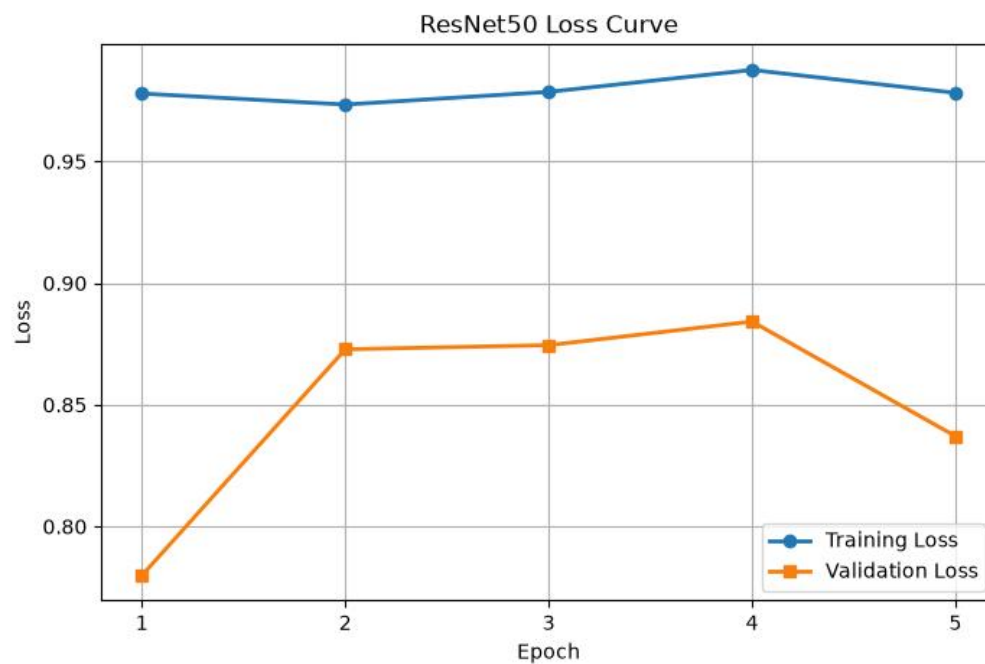
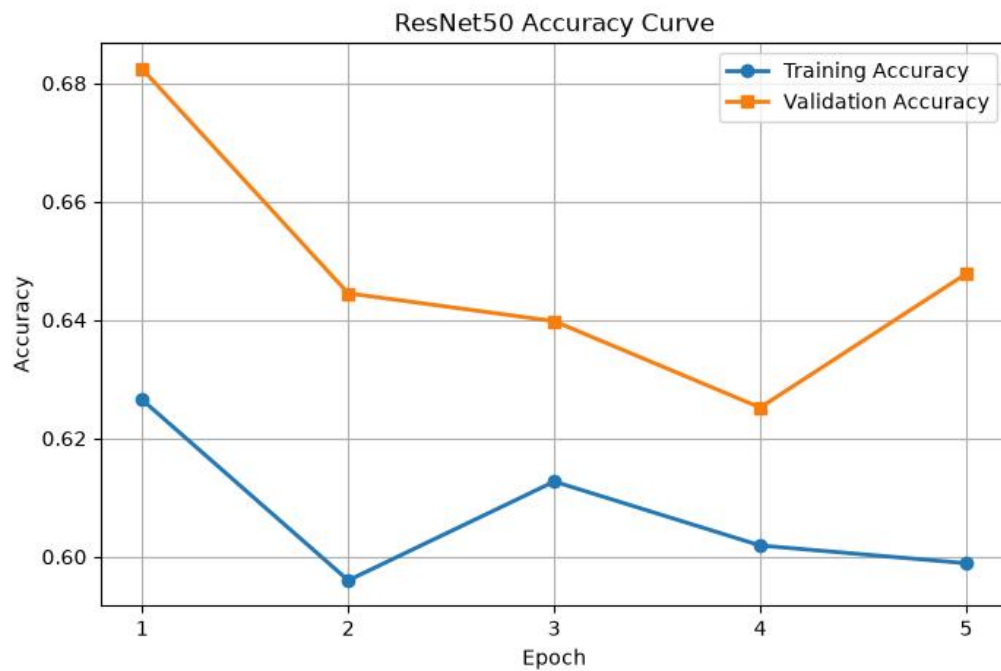
Model Performance Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
ResNet50	0.677977	0.769378	0.677977	0.705844	0.919638
EfficientNet	0.662675	0.803758	0.662674	0.698066	0.931345
DenseNet121	0.650033	0.760364	0.650033	0.686857	0.896458
Deep CNN	0.675316	0.496052	0.675316	0.556741	0.805972
CNN + Dropout	0.671324	0.460758	0.671324	0.541671	0.827283
Basic CNN	0.616766	0.697328	0.616766	0.646965	0.868299
CNN + Batch Normalization	0.616766	0.769448	0.616766	0.660176	0.903196
MobileNetV2	0.590818	0.725034	0.590818	0.629566	0.8533

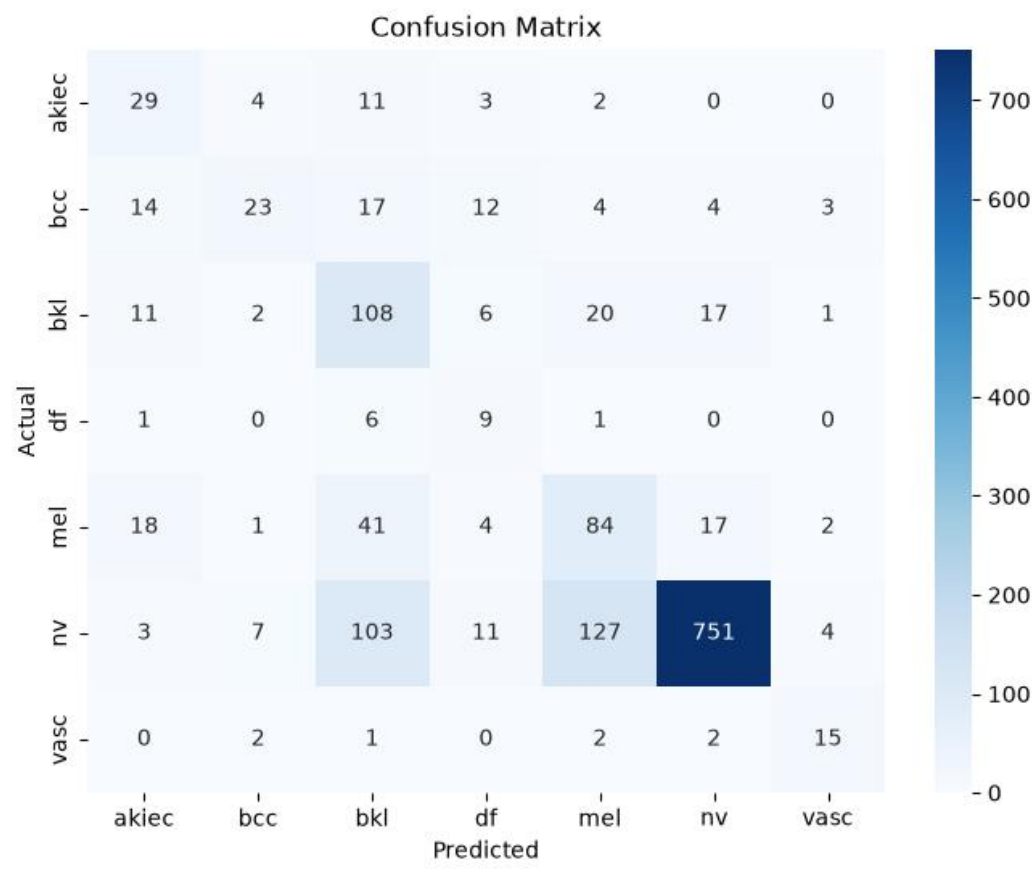
Performance Analysis

The performance of all trained models was compared using Accuracy, Precision, Recall, F1-Score, and ROC-AUC. The results indicate that transfer learning models generally outperformed the custom CNN models.

Among all the evaluated models, **ResNet50** achieved the highest overall performance with an **Accuracy of 67.80%** and an **F1-Score of 70.58%**. Although **EfficientNet** recorded the highest **Precision (80.38%)** and **ROC-AUC (93.13%)**, **ResNet50** provided a better balance across all evaluation metrics. Therefore, **ResNet50 with the SGD optimizer** was selected as the final model for Explainable AI analysis and deployment in the Flask web application.



Resnet Confusion Matrix



Summary

The comparative analysis demonstrates that transfer learning models achieved better classification performance than the custom CNN models. Based on the evaluation metrics, **ResNet50 (SGD)** was identified as the best-performing model and was chosen as the final model for the proposed skin disease classification system.